

16 methods and comparators over 58 evaluations; blue clears that column's floor, orange falls below.

	cell-context												donor				unseen perturbation						
	T1 · Kang									T5c · OP3			T2	T3 · CRISPR					T4	T5u			
ctrl-pred	-.02	.00	.02	.00	.01	-.02	.01	.00	-.03	.01	-.01	.01	.00										
cell-mean floor	.67	.62	.62	.71	.53	.46	.53	.58	.07	.02	.00	.01	.26										
donor-shift	.83	.79	.72	.82	.60	.82	.88	.78	.28	.19	.26	.32	.02										
linear-PCA floor	.83	.78	.72	.82	.60	.82	.88	.78	.30	.18	.26	.33	.04										
scGen	.77	.78	.77	.73	.54	.84	.82	.75	.13	.16	.24	.19	.15										
CPA/chemCPA	.36	.34	.53	.50	.27	.41	.26	.41	.11	-.02	.10	.04	.21										
biolord																							
GEARS																							
AttnPert																							
scGPT	.67	.64	.62	.69	.54	.40	.54	.58	.38	.28	.35	.38	.25										
scFound.	.75	.73	.68	.70	.52	.69	.69	.65	.28	.28	.38	.38	.12										
STATE	.59	.63	.68	.70	.41	.76	.79	.58	.08	.00	.03	.05	.14										
PertAdapt																							
CellOT	.81	.82	.76	.80	.56	.91	.90	.74	.15	.15	.26	.23	.37										
scPRAM	.80	.73	.69	.81	.54	.90	.86	.68	.19	.12	.22	.20	.16										
PerturbNet	.79	.79	.78	.82	.55	.94	.88	.75	.12	.10	.22	.15	.25										
PRnet									.13	.04	.08	.13											
CellFlow	.83	.82	.79	.83	.56	.93	.88	.77	.10	.07	.09	.15	.24										
FP-ridge									.42	.31	.39	.44											
lin-shift-KO																							
	B	CD4T	CD8T	DC	Mk	Mono14	Mono16	NK	B	Mono	NK	T	Soskic										
	Chen	McCutcheon-a	McCutcheon-i	Schmidt	Shifrut	Frangieri	OP3																

cell fill = margin over that column's binding floor (Pearson-Δ)

★ clears both floor members and survives multiplicity correction
 ★ point-estimate clearance, without statistical support

blue = above the floor; white = at the floor; orange = below it

 adapted interface
 diagnostic comparator
 not evaluated

row accent: baseline
 conditioned
 diagnostic comparators

93 of 106 donors
over the cell-mean floor
mean gap +0.107 [+0.086, +0.128]
paired Wilcoxon $p = 1.6 \times 10^{-13}$

mean +0.107

cell-mean floor

(Pearson- Δ)

donor (Soskic CD4 leave-one-donor-out), sorted by gap